

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1074

Indian Hospital 138 kV
February 21, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1074. The Interconnection Customer proposes a 214.1 MW solar photovoltaic facility interconnecting at the Indian Hospital 138 kV substation in the West Virginia area. The request was received on October 09, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 373 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$157,364,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Carolands-Booster 138.0 kV
Contingency: Goosport-Louisiana Pacific 138.0 kV
Loading: 168.8% (Rating: 210 MVA)
DFAX: 18.3%
Pre-project loading: 150.1%
2. Facility: Novato-Carolands 138.0 kV L
Contingency: Goosport-Louisiana Pacific Substation 138.0 kV
Loading: 159.3% (Rating: 220 MVA)
DFAX: 18.3%
Pre-project loading: 141.4%
3. Facility: Border Steel-Riverton 115 138.0 kV
Contingency: Marquette Energy Ctr-Huston 138.0 kV Line
Loading: 155.5% (Rating: 208 MVA)
DFAX: 31.4%
Pre-project loading: 123.2%
4. Facility: Riverton 115-Spencer Gas Turbine 138.0 kV L
Contingency: Marquette Energy Ctr-Huston 138.0 kV line
Loading: 155.1% (Rating: 211 MVA)
DFAX: 31.4%
Pre-project loading: 123.1%
5. Facility: Spencer Gas Turbine-Cedar Crest 138.0 kV
Contingency: Huston-Maquette Energy Ctr 138.0 kV
Loading: 154.5% (Rating: 212 MVA)
DFAX: 31.4%
Pre-project loading: 122.9%

6. Facility: Louisiana Pacific-Cedar Crest 138.0 kV
Contingency: Marquette Energy Ctr-Huston Substation 138.0 kV Line
Loading: 152.7% (Rating: 217 MVA)
DFAX: 31.4%
Pre-project loading: 121.7%
7. Facility: Louisiana Pacific-Goosport 138.0 kV line
Contingency: Marquette Energy Center-Huston Substation 138.0 kV
Loading: 148.2% (Rating: 225 MVA)
DFAX: 31.4%
Pre-project loading: 118.4%
8. Facility: Millbrae-Chester Generating Sta 138.0 kV L
Contingency: Border Steel-Halsey 138.0 kV Line
Loading: 141.1% (Rating: 212 MVA)
DFAX: 14.3%
Pre-project loading: 126.7%
9. Facility: Marquette Energy Ctr-Booster 138.0 kV
Contingency: Louisiana Pacific-Goosport 138.0 kV
Loading: 140.1% (Rating: 223 MVA)
DFAX: 29.4%
Pre-project loading: 112.0%
10. Facility: Huston-Marquette Energy Center 138.0 kV L
Contingency: Louisiana Pacific-Goosport 138.0 kV
Loading: 139.5% (Rating: 228 MVA)
DFAX: 29.4%
Pre-project loading: 111.9%
11. Facility: Indian Hospital-Grantsdale 138.0 kV line
Contingency: Pana N-Rec Harmony Substation 138.0 kV
Loading: 136.4% (Rating: 231 MVA)
DFAX: 63.6%
Pre-project loading: 77.6%
12. Facility: Goosport-Grantsdale 138.0 kV line
Contingency: Paa N-Rec Harmony Substation 138.0 kV
Loading: 136.0% (Rating: 231 MVA)
DFAX: 63.6%
Pre-project loading: 77.1%
13. Facility: Halsey Substation-Lodi 138.0 kV Line
Contingency: Chester Generating Sta-Millbrae Substation 138.0 kV
Loading: 125.7% (Rating: 206 MVA)
DFAX: 16.8%
Pre-project loading: 108.3%
14. Facility: Halsey-Border Steel 138.0 kV
Contingency: Millbrae-Chester Generating Sta 138.0 kV
Loading: 125.2% (Rating: 211 MVA)
DFAX: 16.8%
Pre-project loading: 108.1%
15. Facility: Lodi-Morgantown Energy Facility 138.0 kV
Contingency: Chester Generating Sta-Millbrae Substation 138.0 kV line
Loading: 118.9% (Rating: 194 MVA)
DFAX: 16.8%
Pre-project loading: 100.4%
16. Facility: Huntersville Delivery No 2-Mahoning 138.0 kV
Contingency: Indian Hospital-Grantsdale 138.0 kV

- Loading: 110.3% (Rating: 199 MVA)
DFAX: 17.1%
Pre-project loading: 91.9%
17. Facility: Wareham Valley-Borer Steel 138.0 kV
Contingency: Halsey-Border Steel 138.0 kV
Loading: 107.7% (Rating: 184 MVA)
DFAX: 21.3%
Pre-project loading: 83.0%
18. Facility: Cole-Pleasant Vly Mn 138.0 kV L
Contingency: Indian Hospital-Grantsdale 138.0 kV
Loading: 106.5% (Rating: 235 MVA)
DFAX: 17.1%
Pre-project loading: 90.9%
19. Facility: Huston-Goosport 138.0 kV
Contingency: Goosport Substation-Louisiana Pacific 138.0 kV line
Loading: 106.0% (Rating: 184 MVA)
DFAX: 61.8%
Pre-project loading: 34.0%
20. Facility: Pleasant Vly Mn-Millers Ferry 138.0 kV
Contingency: Indian Hospital-Grantsdale 138.0 kV line
Loading: 105.5% (Rating: 247 MVA)
DFAX: 17.1%
Pre-project loading: 90.7%
21. Facility: Middleburg-Millers Ferry 138.0 kV
Contingency: Indian Hospital-Grantsdale 138.0 kV line
Loading: 105.4% (Rating: 251 MVA)
DFAX: 17.1%
Pre-project loading: 90.7%
22. Facility: Middleburg-Revere Co 138.0 kV
Contingency: Indian Hospital Substation-Grantsdale 138.0 kV
Loading: 104.8% (Rating: 255 MVA)
DFAX: 17.1%
Pre-project loading: 90.4%
23. Facility: Revere Co-Haynes 138.0 kV
Contingency: Revere Co-Ormond 138.0 kV
Loading: 102.4% (Rating: 176 MVA)
DFAX: 5.2%
Pre-project loading: 96.1%
24. Facility: Oasis-Wareham Valley 138.0 kV Line
Contingency: Halsey Substation-Border Steel 138.0 kV Line
Loading: 101.8% (Rating: 191 MVA)
DFAX: 21.3%
Pre-project loading: 78.0%
25. Facility: Haynes-Sarno 138.0 kV L
Contingency: Ormond-Revere Co 138.0 kV
Loading: 100.9% (Rating: 182 MVA)
DFAX: 5.2%
Pre-project loading: 94.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carolands-Booster 138.0 kV	Rebuild line with higher-capacity conductor	\$5,530,000	90% (\$4,969,000)
Novato-Carolands 138.0 kV L	Rebuild line with higher-capacity conductor	\$10,027,000	90% (\$9,009,000)
Riverton 115-Border Steel 138.0 kV	Rebuild line with higher-capacity conductor	\$5,198,000	94% (\$4,914,000)
Spencer Gas Turbine-Riverton 115 138.0 kV line	Rebuild line with higher-capacity conductor	\$13,028,000	94% (\$12,315,000)
Spencer Gas Turbine-Cedar Crest 138.0 kV	Rebuild line with higher-capacity conductor	\$6,006,000	94% (\$5,678,000)
Louisiana Pacific-Cedar Crest 138.0 kV	Rebuild line with higher-capacity conductor	\$11,217,000	94% (\$10,604,000)
Goosport-Louisiana Pacific 138.0 kV	Rebuild line with higher-capacity conductor	\$9,978,000	94% (\$9,432,000)
Millbrae-Chester Generating Sta 138.0 kV	Rebuild line with higher-capacity conductor	\$17,025,000	76% (\$12,956,000)
Marquette Energy Ctr-Booster 138.0 kV	Rebuild line with higher-capacity conductor	\$10,270,000	95% (\$9,719,000)
Huston-Marquette Energy Center 138.0 kV L	Rebuild line with higher-capacity conductor	\$16,933,000	95% (\$16,025,000)
Indian Hospital-Grantsdale 138.0 kV Line	Rebuild line with higher-capacity conductor	\$14,205,000	95% (\$13,468,000)
Goosport-Grantsdale 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,427,000	95% (\$6,093,000)
Halsey-Lodi 138.0 kV	Rebuild line with higher-capacity conductor	\$19,381,000	75% (\$14,521,000)
Halsey-Border Steel 138.0 kV	Rebuild line with higher-capacity conductor	\$5,759,000	75% (\$4,315,000)
Lodi Substation-Morgantown Energy Facility 138.0 kV	Rebuild line with higher-capacity conductor	\$1,362,000	75% (\$1,021,000)
Huntersville Delivery No 2-Mahoning Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$3,507,000	78% (\$2,740,000)
Wareham Valley-Border Steel 138.0 kV	Reconductor line	\$1,573,000	100% (\$1,573,000)
Cole-Pleasant Vly Mn 138.0 kV	Reconductor line	\$4,408,000	78% (\$3,444,000)
Goosport-Huston 138.0 kV	Reconductor line	\$4,451,000	95% (\$4,223,000)
Pleasant Vly Mn-Millers Ferry 138.0 kV	Reconductor line	\$2,414,000	78% (\$1,886,000)
Middleburg-Millers Ferry 138.0 kV	Reconductor line	\$2,231,000	78% (\$1,743,000)
Middleburg-Revere Co 138.0 kV	Reconductor line	\$2,560,000	78% (\$2,000,000)
Haynes-Revere Co 138.0 kV line	Reconductor line	\$1,970,000	67% (\$1,321,000)
Oasis-Wareham Valley 138.0 kV Line	Reconductor line	\$1,570,000	100% (\$1,570,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Haynes-Sarno 138.0 kV L	Reconductor line	\$2,721,000	67% (\$1,825,000)
	Total allocated to project		\$157,364,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.